Quantification: 876_z82c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	8
interior lacuna count	6
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 6)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1771.5	1867	619.765	px^2
major_axis_length_px	lacuna shape	75.501	77.7477	19.5694	px
minor_axis_length_px	lacuna shape	30.3022	30.7562	5.812	px
aspect_ratio	lacuna shape	2.5004	2.4623	0.504	unitless
eccentricity	lacuna shape	0.9052	0.9137	0.0454	unitless
solidity	lacuna shape	0.8535	0.8701	0.0443	unitless
roots_count	headline	9.1667	8.5	4.0702	count
ring_length_r30_px	headline	647	639	121.156	px
ring_length_r60_px	reported	1740.33	1824	333.085	px
owned_length_px	ownership-dependent	7212.99	7497.81	3116.77	px
edge_count	ownership-dependent	529	552.5	219	count
mean_edge_length_px	ownership-dependent	13.4933	13.3259	0.6362	px
perimeter_px	normalised	233.197	239.049	60.4736	px
ring_area_r30_px2	normalised	8429.67	8550	1388.13	px^2
ring_area_r60_px2	normalised	21961.5	22639.5	3572.96	px^2
in_frame_fraction_r30	normalised	0.9983	1	0.0042	unitless
in_frame_fraction_r60	normalised	0.9757	1	0.0596	unitless
ring_density_r30	normalised	0.0770569	0.0779741	0.00968897	px^-1
ring_density_r60	normalised	0.0791821	0.0806921	0.00775523	px^-1
roots_per_100px_perimeter	normalised	3.8628	4.0513	1.089	per 100 px
ring_attached_length_r30_px	network v2	619.667	601.5	116.065	px
ring_attached_length_r60_px	network v2	1539.33	1650.5	358.692	px
ring_length_w_r30_px	network v2	732.684	726.165	133.474	px
ring_length_w_r60_px	network v2	1964.86	2068.71	374.323	px
sholl_crossings_r10	network v2	15.5	16.5	3.2711	count
sholl_crossings_r20	network v2	18.5	19.5	5.1672	count
sholl_crossings_r30	network v2	20.8333	20.5	3.3116	count
loop_count	loops, report only	86	93.5	46.4155	count
loop_edge_share	loops, report only	0.5301	0.5721	0.1401	share
ring_width_mean_r30_px	width	6.5359	6.6556	0.7508	px

Per field
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measure	value	units
analysed_area_px2	1.0316e+06	px^2
total_skeleton_length_px	69561	px
canalicular_length_density_per_px	0.0674305	px^-1
junction_count	2467	count
junction_density_per_px2	0.00239144	px^-2
lacuna_count	8	count

lacunae_per_px2	7.75e-06	px^-2
skeleton_component_count	639	count
mean_component_length_px	108.859	px
median_component_length_px	12	px
field_length_density_w_per_px	0.0757537	px^-1
field_density_without_flagged_per_px	0.067209	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	147	px
unattached_component_count	9	count
canalicular_length_density_attached_per_px	0.0674305	px^-1
field_length_density_with_unattached_per_px	0.067573	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	748	count
loop_edge_share	0.4797	share
edge_count	5378	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.168	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	56385	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	4013	213.469	32.7614	6.5158	0.9882	0.5369	1.527	16.3571	878.108	True
2	2473	101.199	32.7068	3.0941	0.9463	0.7952	1.5403	149.487	378.777	False
3	2169	83.3807	34.4381	2.4212	0.9107	0.8757	-1.5585	303.736	321.658	False
4	1583	72.1147	28.8056	2.5035	0.9168	0.876	-1.5035	531.782	196.779	False
5	1826	64.8209	36.821	1.7604	0.823	0.9062	-1.5669	541.512	442.987	False
6	1908	86.6787	28.6695	3.0234	0.9437	0.8645	-1.4722	691.032	607.199	False
7	670	44.8117	20.3719	2.1997	0.8907	0.8034	-1.4095	908.587	49.1806	False
8								188.435	951.62	True

Network per lacuna

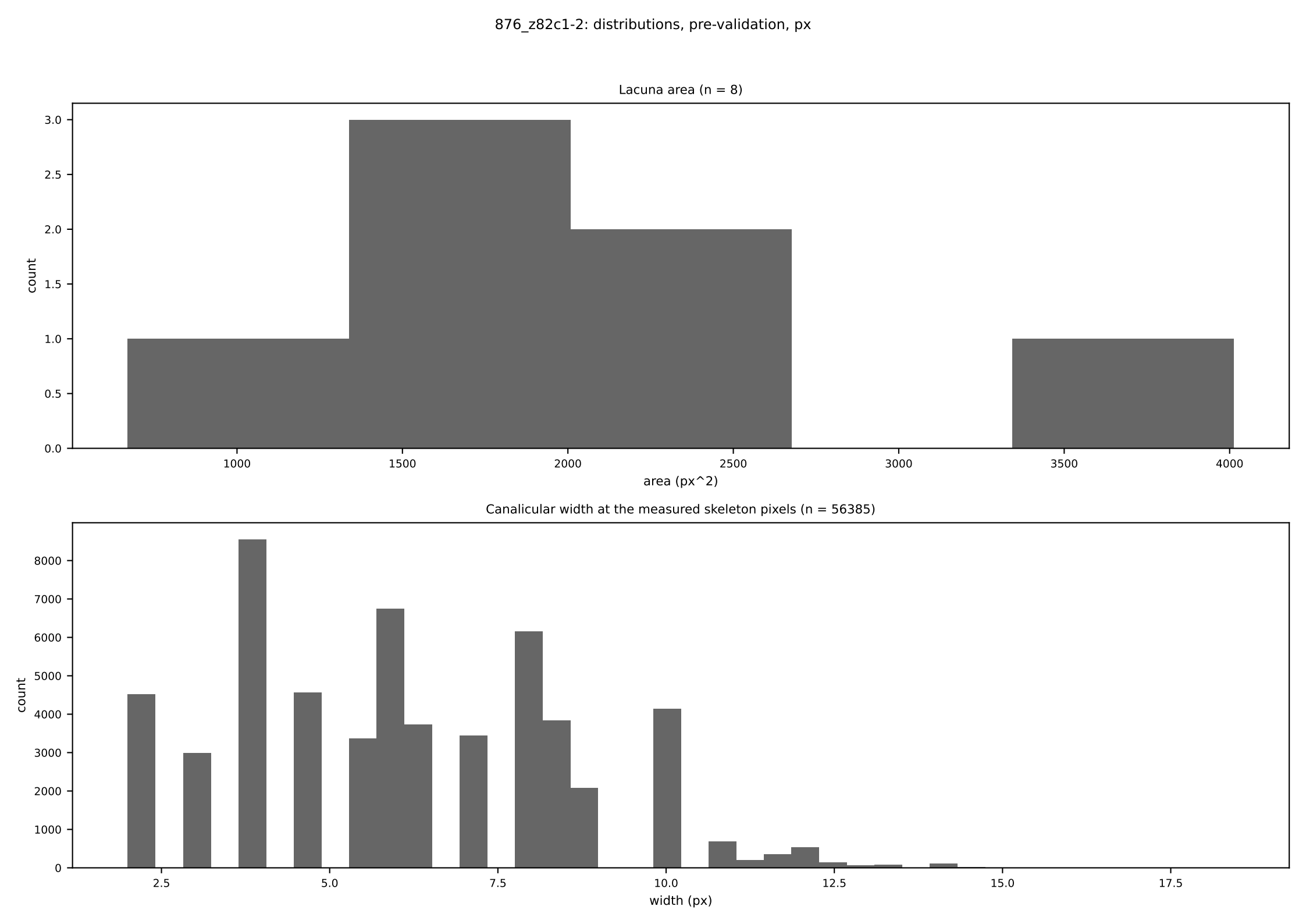
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	26	1091	1899	4505.06	331	13.6105	3
2	15	735	2029	10522.7	724	14.5341	7.6575
3	12	824	2044	8360.33	626	13.3552	6.5752
4	11	633	1814	10039	755	13.2967	6.8577
5	6	645	1834	6635.29	479	13.8524	6.736
6	6	562	1556	5518.69	416	13.2661	5.6855
7	5	483	1165	2202	174	12.6552	5.7038
8	25	1154	2208	8677.91	623	13.9292	7.1441

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	782.839	12986	23176	0.6542	0.5574	0.0840136	0.0819382	3.3212
2	318.718	10329	26001	1	1	0.0711589	0.0780355	4.7064
3	259.262	9043	23619	1	1	0.0911202	0.0865405	4.6285
4	218.836	8099	21764	1	1	0.0781578	0.0833486	5.0266
5	196.936	7897	21388	1	1	0.0816766	0.085749	3.0467
6	261.505	9001	23515	1	1	0.0624375	0.0661705	2.2944
7	143.924	6209	15482	0.9898	0.854	0.0777903	0.0752487	3.4741
8	919.704	16560	31079	0.897	0.8325	0.069686	0.0710448	2.7183

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	1054	1819	1221.34	2116.38	24	26	17
2	698	1770	820.056	2282.39	19	22	23
3	801	1945	933.198	2303.82	17	25	26
4	601	1677	719.973	2055.42	12	19	21
5	602	1624	732.357	2082.01	18	20	18
6	545	1234	635.902	1744.33	16	14	17
7	471	986	554.617	1321.19	11	11	20
8	1053	2012	1274.82	2439.83	32	23	26


876_z82c1-2: distributions, pre-validation, px
Lacuna area (n = 8)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 56385)
count
width (px)